

Lecture 10: NP-completeness: Subset-sum

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School of Computer and Communication Sciences

Lecture 10

Recap: NP-completeness reductions

NP-completeness

Definition: A language L is said to be **NP-complete** if

- ▶ L is in **NP**.
- ▶ For every language L' in **NP**, $L' \leq_P L$.

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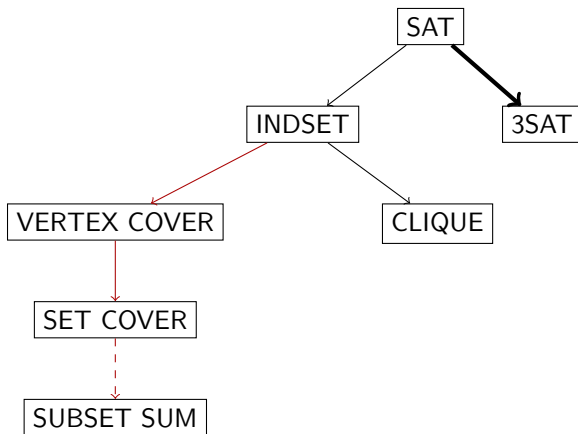
- ▶ L is in **NP**.
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- ▶ **[NP hardness]** Show $C \leq_P L$ for some **NP**-complete language C
(not the other way around)



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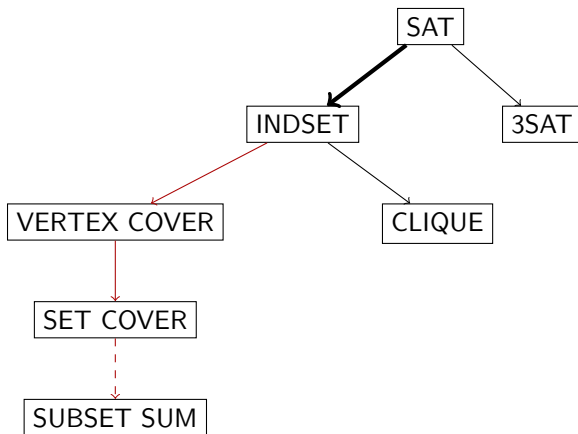
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What is the runtime?

Does $\text{SAT} \leq_P 2\text{SAT}$ analogously?



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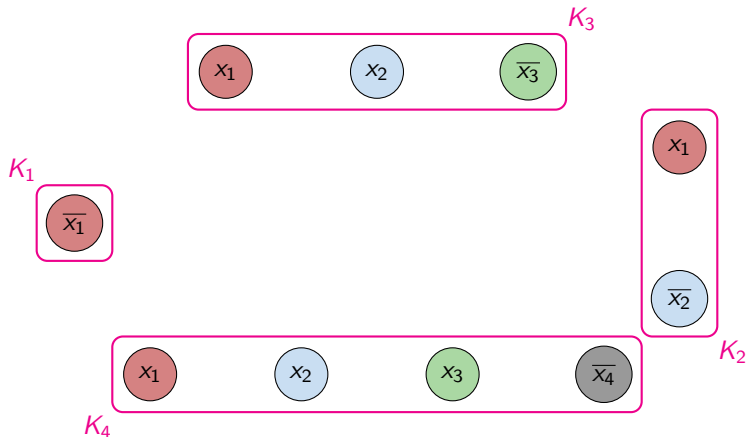
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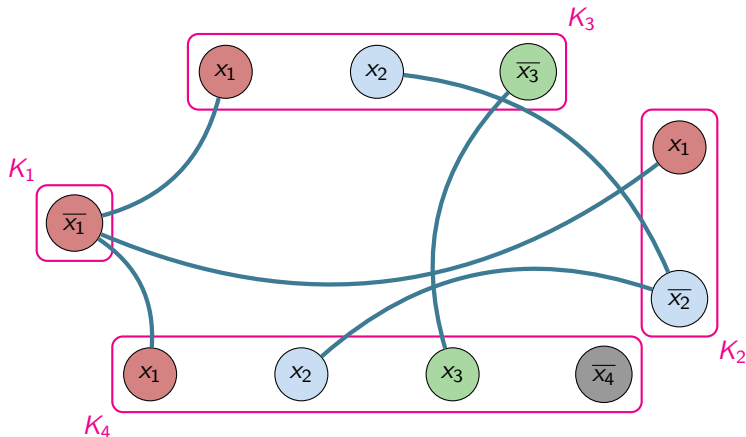
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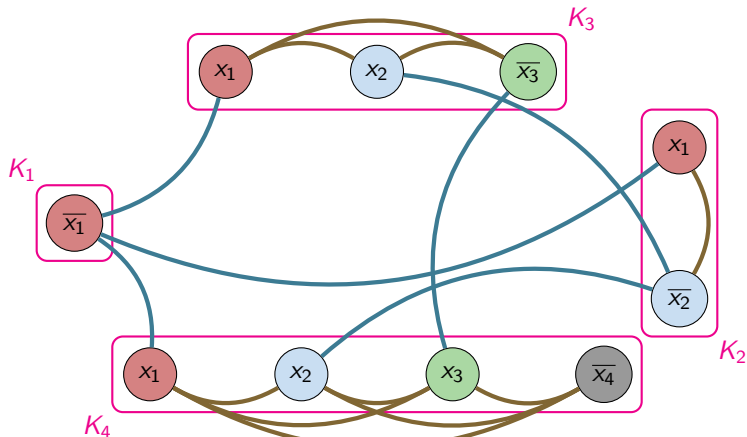
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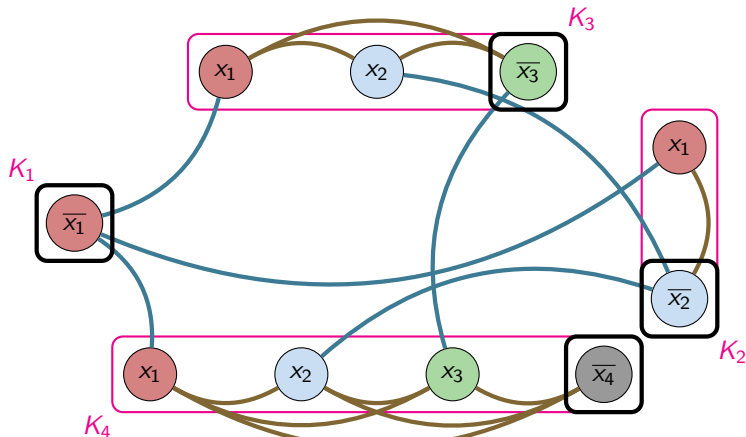
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Proof: C : satisfying assignment of φ . Pick one true literal from each clause. The corresponding vertices form an independent set.

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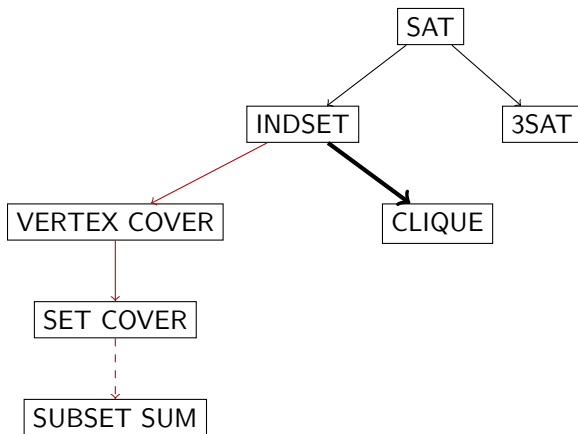
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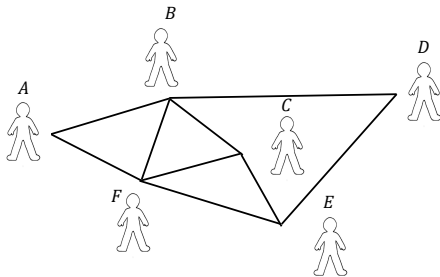
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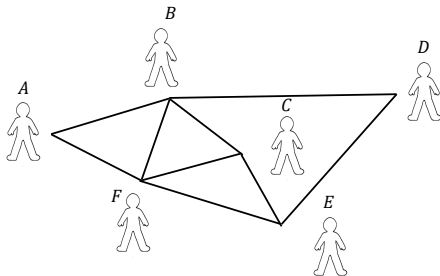
Proof: C : independent set in G , $|C| = m$. C contains one vertex from each group. Set the corresponding literals to true to get a satisfying assignment.



CLIQUE



CLIQUE

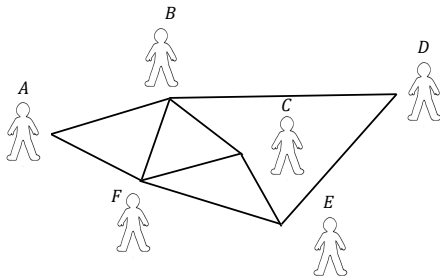


A, B, F is a 3-clique

Definition: A k -clique is a subset of k pairwise connected vertices

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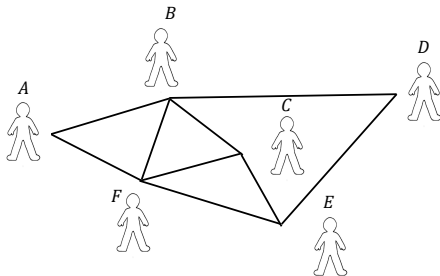
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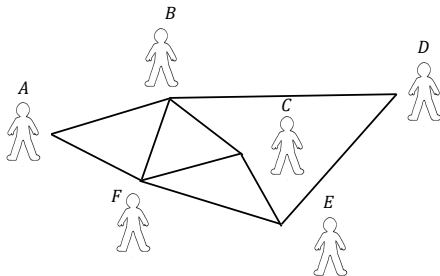
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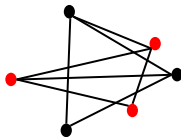
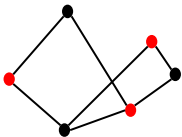
$\langle G, 4 \rangle \in \text{CLIQUE}$? No

INDSET vs CLIQUE

For a graph G :

Independent set: A subset of pairwise **non-adjacent** vertices

Clique: A subset of pairwise **adjacent** vertices

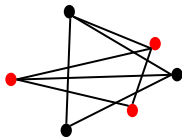
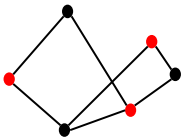


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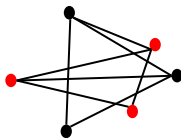
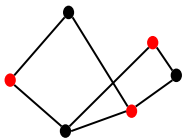
Def (Complement): The *complement* of a graph $G = (V, E)$ is a graph $\bar{G} = (V, \bar{E})$ with same vertex set and edge set $\bar{E}$ s.t. $uv \in \bar{E}$ iff $uv \notin E$

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Observation: If G is a graph and $\bar{G}$ its complement, then a subset S of the vertices of G is an independent set iff S is a clique of $\bar{G}$

INDSET $\leq_p$ CLIQUE

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Correctness: $\langle G, k \rangle \in \text{INDSET} \Leftrightarrow \langle \bar{G}, k \rangle \in \text{CLIQUE}$

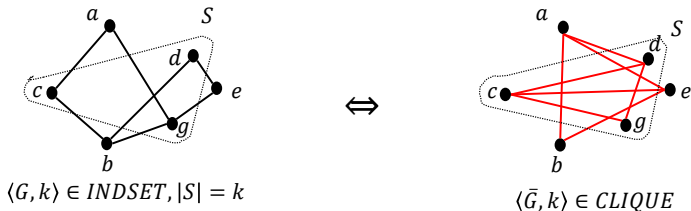
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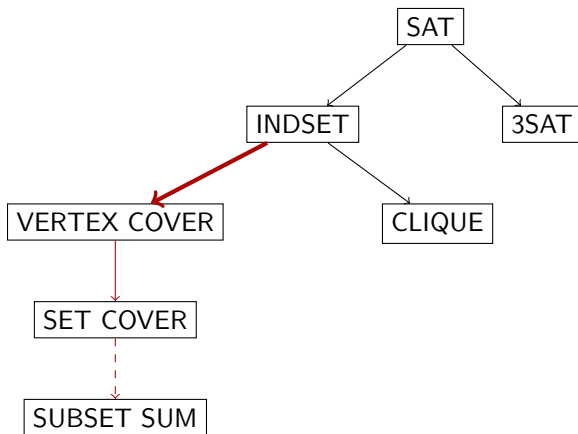
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Proof of correctness:

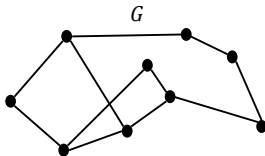




Vertex Cover

Definition: For a graph $G = (V, E)$, a *vertex cover* is a subset S of V such that every edge of G is incident to at least one vertex in S

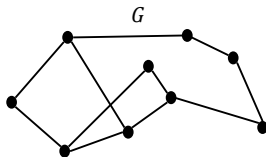
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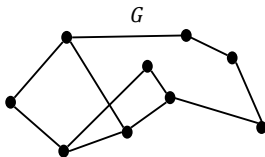
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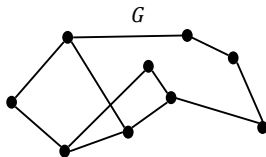
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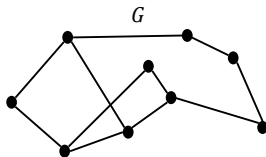
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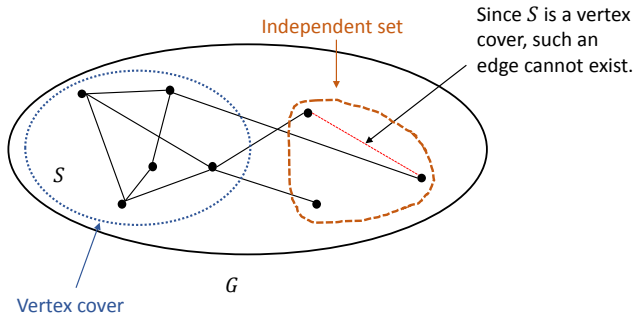
INDSET vs VERTEX COVER

INDSET vs VERTEX COVER

Lemma: For every graph G , a subset S of the vertices is a vertex cover if and only if $\bar{S}$ is an independent set where $\bar{S} = V \setminus S$

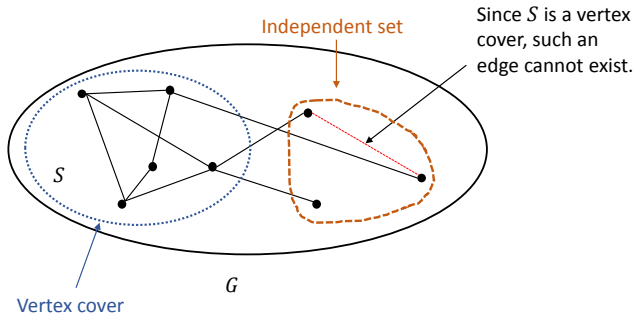
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Corollary: For every graph G and a positive integer k , G has an independent set of size k iff G has a vertex cover of size $n - k$

INDSET $\leq_p$ VERTEX COVER

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Reduction: $f(\langle G, k \rangle) := \langle G, n - k \rangle$, where n is the number of vertices of G

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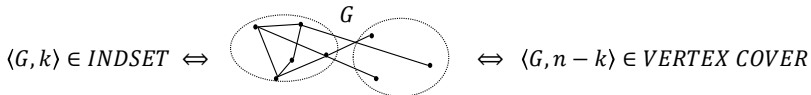
Efficiency: The reduction f is polynomial time

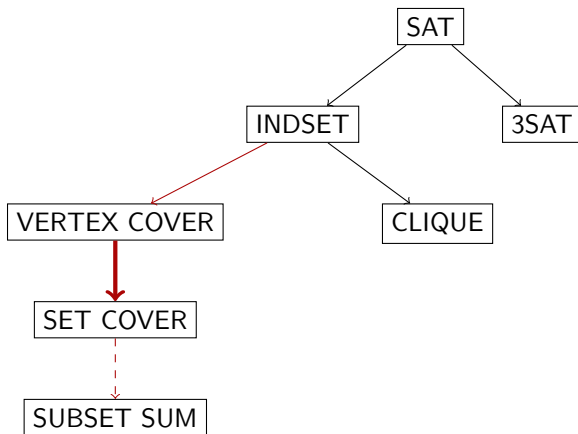
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Correctness: Lemma/Corollary on previous slide!

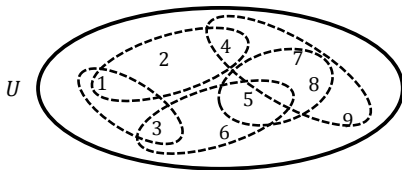




Set Cover

Definition: Let $U = \{1, \dots, n\}$ and $\mathcal{F} = \{T_1, \dots, T_m\}$ be a family of subsets $\forall i, T_i \subseteq U$

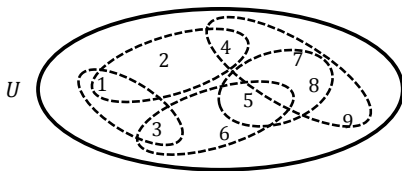
A subset $\{T_{i_1}, \dots, T_{i_k}\} \subseteq \mathcal{F}$ is called a **set cover** of size k if $\bigcup_{j=1}^k T_{i_j} = U$



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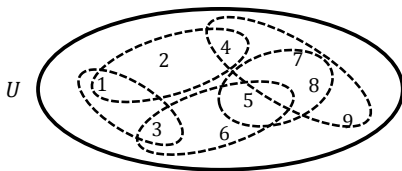
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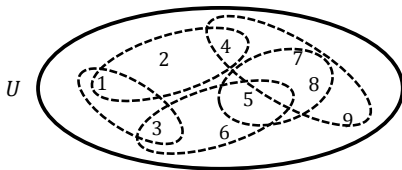
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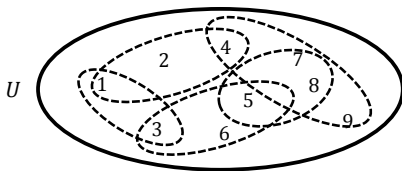
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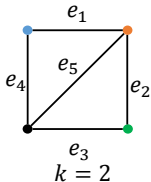
VERTEX COVER $\leq_p$ SET COVER

Key idea: VERTEX COVER is a *special case* of SET COVER

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Example:



$$U = \{e_1, e_2, e_3, e_4, e_5\}$$

$$\mathcal{F} = \{\{e_1, e_4\}, \{e_1, e_2, e_5\}, \\ \{e_2, e_3\}, \{e_3, e_4, e_5\}\}$$

$$k' = k = 2$$

VERTEX COVER $\leq_p$ SET COVER

Key idea: VERTEX COVER is a *special case* of SET COVER



Reduction:

- ▶ Let $\langle G = (V, E), k \rangle$ be an instance of VERTEX COVER
- ▶ Set $U := E$
- ▶ For every vertex $v \in V$ create a set S_v

$$S_v := \{e \in E \mid e \text{ is incident to } v\}$$

- ▶ Let $\mathcal{F} := \{S_v \mid v \in V\}$ and set $k' = k$

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Efficiency: Obvious from construction

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- ▶ Suppose $C \subseteq V$ is a vertex cover of G and $|C| = k$
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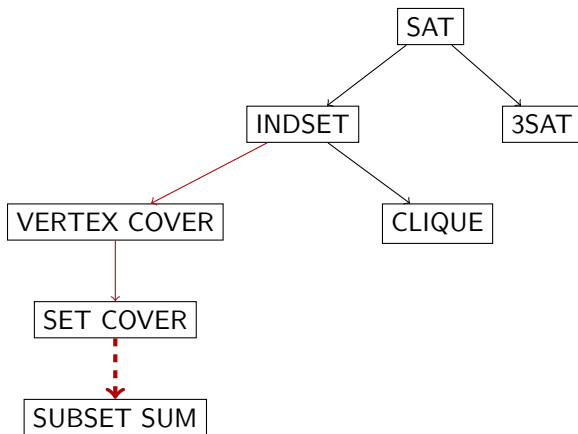
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Correctness ($\Leftarrow$): If U can be covered by k sets, then G has a vertex cover of cardinality k

Proof:

- ▶ Let $S_{v_1}, S_{v_2}, \dots, S_{v_k}$ be a collection of sets which cover $U = E$
- ▶ We claim that $C = \{v_1, v_2, \dots, v_k\}$ is a vertex cover of G
- ▶ Indeed, every edge e in G belongs to S_{v_i} for some $i \in \{1, 2, \dots, k\}$
- ▶ Hence, every edge e in G is incident to some vertex $v_i \in C$



SUBSET-SUM

Let X denote a (multi) set of positive integers

Definition: SUBSET-SUM

$= \{ \langle X, s \rangle : X \text{ contains a subset whose elements sum to } s \}$

Example: $X = \{1, 3, 4, 6, 13, 13\}$

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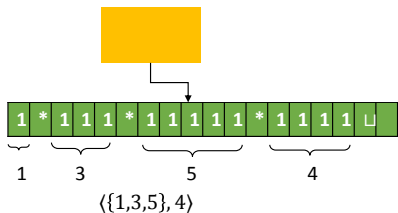
It depends on the **input length**

In *SUBSET-SUM*
we use binary
encoding

Unary

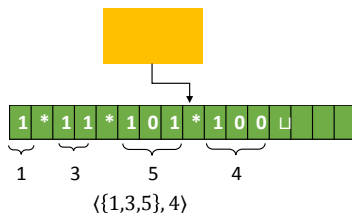
vs

Binary



Input length: $x_1 + x_2 + \dots + x_m + s$

>>



Input length: $\lceil \log(x_1) \rceil + \dots + \lceil \log(x_m) \rceil + \lceil \log(s) \rceil$

SUBSET-SUM: An Algorithm

- Let $X = \{x_1, x_2, \dots, x_m\}$, $sum := \sum_{j=1}^m x_j$
- FOR $i = 1 \dots m$ DO
- FOR $j = 0 \dots sum$ DO
- $A[i, j] := False$
- FOR $i = 0$ to m DO
- $A[i, 0] := True$
- FOR $i = 1 \dots m$ DO
- FOR $j = 1 \dots sum$ DO
- IF $A[i - 1, j - x_i] = True$ OR $A[i - 1, j] = True$ THEN $A[i, j] := True$
- IF $A[m, sum] = True$, ACCEPT; ELSE, REJECT

Running time: For the input $\langle \{x_1, \dots, x_m\}, s \rangle$, the above algorithm takes $O(m \cdot (\sum_{i=1}^m x_i))$ steps to output whether or not $\langle X, s \rangle \in SUBSET-SUM$

Unary representation: The length of the input $x_1 + x_2 + \dots + x_m + s$. In this case the running time is **polynomial** in the input length

Binary representation: The length of the input $\lceil \log(x_1) \rceil + \dots + \lceil \log(x_m) \rceil + \lceil \log(s) \rceil$. In this case, the running time is **exponential** in the input length

SUBSET-SUM is **NP**-Complete

Theorem: SET COVER $\leq_p$ SUBSET-SUM

Corollary: SUBSET-SUM is **NP**-Complete

Indeed, SUBSET-SUM $\in$ **NP** as it is easy to verify in polynomial time that $\sum_{x \in S} x = s$ for a given subset $S \subseteq X$.

SET COVER vs SUBSET-SUM

The goal: SET COVER $\leq_p$ SUBSET-SUM

	SET COVER	SUBSET SUM
INPUT:	▶ $U = \{1, 2, \dots, n\}$ ▶ A family $\mathcal{F} = \{T_1, \dots, T_m\}$ of subsets of U ▶ A number k	▶ A set of positive integers $X = \{x_1, \dots, x_\ell\}$ ▶ A positive integer s
GOAL:	Select k elements of $\mathcal{F}$ whose union is U	Select a subset of X whose sum of elements is s

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First Question: How to map subsets to numbers?

Encoding subsets as numbers

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$$U = \begin{array}{|c|c|c|c|c|c|c|} \hline 1 & 2 & 3 & 4 & \dots & n-1 & n \\ \hline \end{array}$$

Encoding subsets as numbers

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$T = \{2, 3, n\}$ is then encoded by

$$T \rightarrow \begin{array}{|c|c|c|c|c|c|c|} \hline 0 & 1 & 1 & 0 & \dots & 0 & 1 \\ \hline \end{array}$$

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$$\begin{array}{l} \text{Binary:} \end{array} \quad \begin{array}{|c|c|c|c|c|c|c|} \hline 2^{n-1} & 2^{n-2} & 2^{n-3} & \dots & 2^1 & 2^0 \\ \hline b_1 & b_2 & b_3 & & b_{n-1} & b_n \\ \hline \end{array} \quad \sum_{j=1}^n b_j 2^{n-j}$$

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c-ary:
(for $c \in \mathbb{N}$)

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Reduction: Part 1 (encoding sets as numbers)

$$\langle U = \{1, \dots, n\}, \mathcal{F} = \{T_1, T_2, \dots, T_m\}, k \rangle \implies X = \{a_1, \dots, a_m\} \text{ in which}$$

$$T_i \implies \begin{array}{|c|c|c|c|} \hline 0 & 1 & \dots & 1 \\ \hline \end{array} \implies \begin{array}{|c|c|c|c|} \hline 0 & 1 & \dots & 1 \\ \hline \end{array}_c \implies a_i = \sum_{j=1}^n b_j c^{n-j}$$

- ▶ We set the value of c later

How to pick s ?

The reduction should ensure that:

$$\langle U, \mathcal{F}, k \rangle \in \text{SET COVER} \Rightarrow \langle X, s \rangle \in \text{SUBSET-SUM}$$

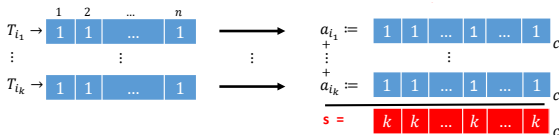
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More generally: $\bigcup_{j=1}^k T_{i_j} = U$ but sets not identical

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 T_{i_k} \rightarrow \begin{array}{|c|c|c|c|c|c|} \hline 1 & 1 & \dots & 0 & \dots & 0 \\ \hline \end{array} & \longrightarrow & a_{i_k} := \begin{array}{|c|c|c|c|c|c|} \hline 1 & 1 & \dots & 0 & \dots & 0 \\ \hline \end{array}_c \\
 & & \hline
 \Sigma_{j=1}^k a_{i_j} = \begin{array}{|c|c|c|c|c|c|} \hline t_1 & t_2 & \dots & t_n & & \\ \hline \end{array}_c \leq \begin{array}{|c|c|c|c|c|c|} \hline k & k & \dots & k & & \\ \hline \end{array}_c
 \end{array}$$

Reduction: Part 2 (selecting s)

$$\begin{array}{ccc}
 \begin{array}{c} T_{i_1} \rightarrow \\ \vdots \\ T_{i_k} \rightarrow \end{array} & \begin{array}{c} \begin{array}{cccccc} 1 & 2 & & i & \dots & n \end{array} \\ \begin{array}{cccccc} 0 & 1 & \dots & 1 & \dots & 1 \\ \vdots & \vdots & & \vdots & & \vdots \\ 1 & 1 & \dots & 0 & \dots & 0 \end{array} \end{array} & \xrightarrow{\quad \quad} & \begin{array}{c} a_{i_1} := \\ \vdots \\ a_{i_k} := \end{array} \begin{array}{c} \begin{array}{cccccc} 0 & 1 & \dots & 1 & \dots & 1 \\ \vdots & \vdots & & \vdots & & \vdots \\ 1 & 1 & \dots & 0 & \dots & 0 \end{array} \\ \hline \begin{array}{cccccc} t_1 & t_2 & \dots & t_h & \dots & t_n \end{array} \end{array} \begin{array}{c} c \\ c \\ c \end{array}
 \end{array}$$

$$\sum_{j=1}^k a_{i_j} = \begin{array}{cccccc} t_1 & t_2 & \dots & t_n \end{array}_c \leq \begin{array}{cccccc} k & k & \dots & k \end{array}_c$$

Observation: Since $\bigcup_{j=1}^k T_{i_j} = U, \forall h, 1 \leq t_h \leq k$

Idea 2 (additional numbers) The reduction adds to X for every $h \in \{1, 2, \dots, n\}$, $k - 1$ integers

$$b_h = \begin{array}{cccccc} 0 & 0 & \dots & 1 & \dots & 0 \end{array}_c$$

$\uparrow$
 h -th pos

By adding $k - t_h$ of each b_h 's, we ensure:

$$\text{if } \bigcup_{j=1}^k T_{i_j} = U \text{ then } \sum_{j=1}^k a_{i_j} + \sum_{h=1}^n (k - t_h) b_h = s$$

What about the other direction?

Suppose: $\langle X, s \rangle \in \text{SUBSET-SUM}$

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Suppose: $\langle X, s \rangle \in \text{SUBSET-SUM}$ Thus $\exists a_{i_1}, \dots, a_{i_p}, \underbrace{b_1, \dots, b_1}_{r_1 \text{ times}}, \dots, \underbrace{b_n, \dots, b_n}_{r_n \text{ times}} \in X$

s.t.

$$\sum_{j=1}^p a_{i_j} + r_1 \cdot b_1 + \dots + r_n \cdot b_n = s = \boxed{k} \boxed{\dots} \boxed{k}_c$$

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Claim: $\forall h \in \{1, \dots, n\}, \exists q \in \{1, \dots, p\}$ s.t. the h -th coordinate of a_{i_q} is 1

Proof:

- ▶ Crucial idea: force **no carry over** by selecting c large enough ($c = m + kn + 1$ larger than the number of summands)
- ▶ The h -th digit of $\sum_{j=1}^p a_{i_j} + r_1 \cdot b_1 + \dots + r_n \cdot b_n$ is k
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Final question: $T_{i_1} \cup \dots \cup T_{i_p} = U \Rightarrow \langle U, \mathcal{F}, k \rangle \in \text{SET COVER?}$

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- ▶ Thus, at least one of the subsets $T_{i_1}, \dots, T_{i_p}$ contains h

Final question: $T_{i_1} \cup \dots \cup T_{i_p} = U \Rightarrow \langle U, \mathcal{F}, k \rangle \in \text{SET COVER?}$ **only if** $p = k$

Reduction part 3

Idea 3 (enforce $p = k$): Introduce a **new coordinate** in the leftmost position of a_i 's, b_i 's and s

$$\begin{array}{lcl} T_i & \begin{array}{|c|c|c|c|} \hline 0 & 1 & \dots & 1 \\ \hline \end{array} & \longrightarrow & \begin{array}{l} a_i := \begin{array}{|c|c|c|c|c|} \hline \textcolor{red}{1} & 0 & 1 & \dots & 0 \\ \hline \end{array}_c \\ \qquad \qquad \qquad \downarrow \text{(j+1)-st position} \\ b_j := \begin{array}{|c|c|c|c|c|c|} \hline \textcolor{red}{0} & 0 & 0 & \dots & 1 & \dots & 0 \\ \hline \end{array}_c \\ \\ s := \underbrace{\begin{array}{|c|c|c|c|c|c|} \hline \textcolor{red}{k} & k & k & \dots & k & \dots & k \\ \hline \end{array}}_{n+1}_c \end{array}$$

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Claim: If $\sum_{j=1}^p a_{ij} + r_1 \cdot b_1 + \dots r_n b_n = s$, then $p = k$

Proof

- ▶ The leftmost digit of $\sum_{j=1}^p a_{ij} + r_1 \cdot b_1 + \dots r_n b_n = s$ is k (base c)
- ▶ The leftmost digit of b_i is 0
- ▶ The leftmost digit of a_{ij} is 1
- ▶ **No carry over** (since the base $c = m + kn + 1$ is large enough)
- ▶ Hence, it must be that $p = k$

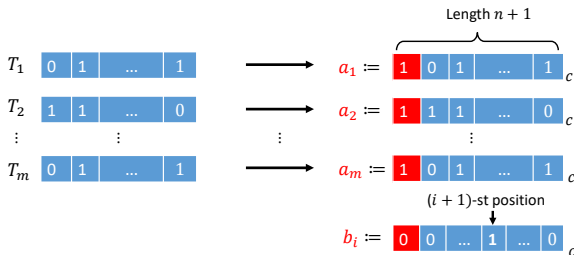
The Complete Reduction

Given: $\langle U = \{1, \dots, n\}, \mathcal{F} = \{T_1, \dots, T_m\}, k \rangle$

Define: $X = \{a_1, \dots, a_m, \underbrace{b_1, \dots, b_1}_{k-1 \text{ times}}, \dots, \underbrace{b_n, \dots, b_n}_{k-1 \text{ times}}\},$

$s := \overbrace{\begin{bmatrix} k & k & k & \dots & k \end{bmatrix}}^{\text{Length } n+1}{}_c$

where:



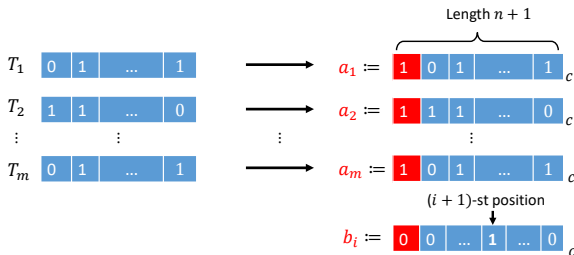
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Theorem: SET COVER $\leq_p$ SUBSET-SUM

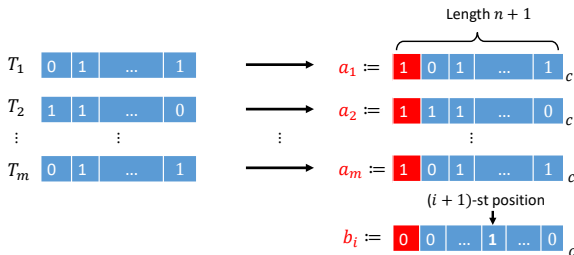
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$s := \overbrace{[k, k, \dots, k]}^{\text{Length } n+1}{}_c$

where:



Theorem: $\text{SET COVER} \leq_p \text{SUBSET-SUM}$

Proof: Efficiency of reduction:

- ▶ the numbers in X we construct have length $\leq (n+1) \log c$ in binary
- ▶ We can compute them in polynomial time

Correctness ($\Rightarrow$)

Suppose: $\langle U, \mathcal{F}, k \rangle \in \text{SET COVER}$

- ▶ Therefore, $\exists T_{i_1}, \dots, T_{i_k}$ s.t. $\cup_{j=1}^k T_{i_j} = U$

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- ▶ Since each element is covered we have $1 \leq t_h \leq k$
- ▶ By adding $(k - t_h)$ copies of b_h we can make the $(h+1)$ st digit equal to k without affecting other digits

Correctness ($\Rightarrow$)

Suppose: $\langle U, \mathcal{F}, k \rangle \in \text{SET COVER}$

- ▶ Therefore, $\exists T_{i_1}, \dots, T_{i_k}$ s.t. $\bigcup_{j=1}^k T_{i_j} = U$
- ▶ First coordinate of $\sum_{j=1}^k a_{i_j}$ is k
- ▶ Let t_h be the $(h+1)$ st coordinate of $\sum_{j=1}^k a_{i_j}$
- ▶ Since each element is covered we have $1 \leq t_h \leq k$
- ▶ By adding $(k - t_h)$ copies of b_h we can make the $(h+1)$ st digit equal to k without affecting other digits
- ▶ Thus, $\sum_{j=1}^k a_{i_j} + (k - t_1)b_1 + \dots + (k - t_n) \cdot b_n = s$

Correctness ($\Rightarrow$)

Suppose: $\langle U, \mathcal{F}, k \rangle \in \text{SET COVER}$

- ▶ Therefore, $\exists T_{i_1}, \dots, T_{i_k}$ s.t. $\bigcup_{j=1}^k T_{i_j} = U$
- ▶ First coordinate of $\sum_{j=1}^k a_{i_j}$ is k
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- ▶ Since each element is covered we have $1 \leq t_h \leq k$
- ▶ By adding $(k - t_h)$ copies of b_h we can make the $(h+1)$ st digit equal to k without affecting other digits
- ▶ Thus, $\sum_{j=1}^k a_{i_j} + (k - t_1)b_1 + \dots + (k - t_n) \cdot b_n = s$
- ▶ Hence, $\langle X, s \rangle \in \text{SUBSET-SUM}$

Correctness ($\Leftarrow$)

Suppose: $\langle X, s \rangle \in \text{SUBSET-SUM}$

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Suppose: $\langle X, s \rangle \in \text{SUBSET-SUM}$ Thus $\exists a_{i_1}, \dots, a_{i_p}, \underbrace{b_1, \dots, b_1}_{r_1 \text{ times}}, \dots, \underbrace{b_n, \dots, b_n}_{r_n \text{ times}} \in X$

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$$\sum_{j=1}^p a_{i_j} + r_1 \cdot b_1 + \dots + r_n \cdot b_n = s = \boxed{k} \boxed{\dots} \boxed{k}_c$$

Let $T_{i_1}, \dots, T_{i_k} \in \mathcal{F}$ be the sets corresponding to $a_{i_1}, \dots, a_{i_k}$

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Step 1: $p = k$

- ▶ There is no carry over when computing the sum
- ▶ The first digit in the sum = number of a_{i_j} 's = p
- ▶ The first digit in $s = k$ and so $p = k$

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- ▶ The $(h+1)$ st digit of $\sum_{j=1}^k a_{i_j} + r_1 \cdot b_1 + \dots + r_n \cdot b_n$ is k
- ▶ The $(h+1)$ st digit of b_i is 1 if $i = h$ and 0 otherwise
- ▶ Since $r_h \leq k = 1$ the $(h+1)$ st digit of $\sum_{j=1}^k a_{i_j}$ is at least 1
- ▶ Therefore the $(h+1)$ st digit of at least one of $a_{i_1}, \dots, a_{i_k}$ is 1
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- ▶ So some set T_{i_q} contains h

Conclusion: $\bigcup_{j=1}^k T_{i_j} = U$ and so $\langle U, \mathcal{F}, k \rangle \in \text{SET COVER}$

NP-complete problems you can use:

CLIQUE, SAT, 3SAT, INDSET
VERTEXCOVER, SETCOVER, SUBSET-SUM,
PERFECT-3-MATCHING